6.2 Customized Ping Program to Measure Min, Max, and Average RTT

AIM:

To develop a Customized Ping Program to Measure Min, Max, and Average RTT

ALGORITHM for Program 2:

- Start.
- Import the required modules.
- Initialize the server address, port number, and number of ping attempts.
- Create an empty list to store RTT values.
- Repeat the following for each ping attempt:
    - Create a TCP socket.
    - Set the timeout value.
    - Record the start time.
    - Connect to the server.
    - Record the end time.
    - Calculate the RTT and store it in the list.
    - Display the RTT.
    - If the connection fails, display **"Request timed out"**.
- Calculate and display:
    - Packets Sent
    - Packets Received
    - Packets Lost
    - Minimum RTT
    - Maximum RTT
    - Average RTT
- Stop.

PROGRAM2: Customized Ping Program to Measure Min, Max, and Average RTT

```
import socket
import time
# Server details
host = "google.com"
port = 80
count = 4
# List to store RTT values
rtt_list = []
print(f"Pinging {host} with TCP packets:\n")
for i in range(count):
    try:
        # Create a TCP socket
        s = socket.socket(socket.AF_INET, socket.SOCK_STREAM)
        s.settimeout(3)
        # Record start time
        start = time.time()
        # Connect to the server
        s.connect((host, port))
        # Record end time
        end = time.time()
        # Close the socket
        s.close()
        # Calculate RTT in milliseconds
        rtt = (end - start) * 1000
        rtt_list.append(rtt)
        print(f"Reply {i+1}: time = {rtt:.2f} ms")
    except Exception:
```

```
        print(f"Reply {i+1}: Request timed out")
# Display statistics
if len(rtt_list) > 0:
    print("\n--- Ping Statistics ---")
    print(f"Packets Sent     : {count}")
    print(f"Packets Received : {len(rtt_list)}")
    print(f"Packets Lost     : {count - len(rtt_list)}")
    print(f"Minimum RTT      : {min(rtt_list):.2f} ms")
    print(f"Maximum RTT      : {max(rtt_list):.2f} ms")
    print(f"Average RTT      : {sum(rtt_list)/len(rtt_list):.2f} ms")
else:
    print("\nNo response received from the server.")
```

SAMPLE OUTPUT:

Pinging google.com with TCP packets:

Reply 1: time = 18.24 ms

Reply 2: time = 17.81 ms

Reply 3: time = 19.15 ms

Reply 4: time = 18.56 ms

--- Ping Statistics ---

Packets Sent   : 4

Packets Received : 4

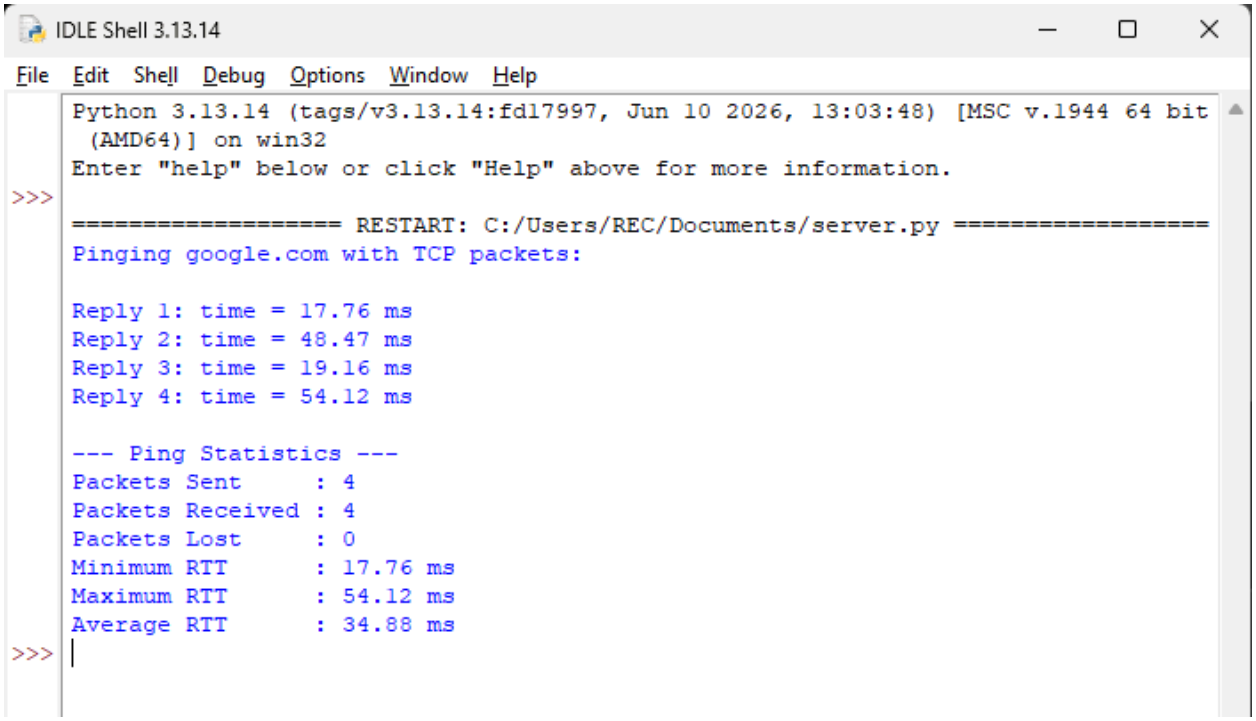
Packets Lost   : 0

Minimum RTT   : 17.81 ms

Maximum RTT   : 19.15 ms

Average RTT   : 18.44 ms

OUTPUT:



RESULT: Therefore a Customized Ping Program to Measure Min, Max, and Average RTT has been created and verified.